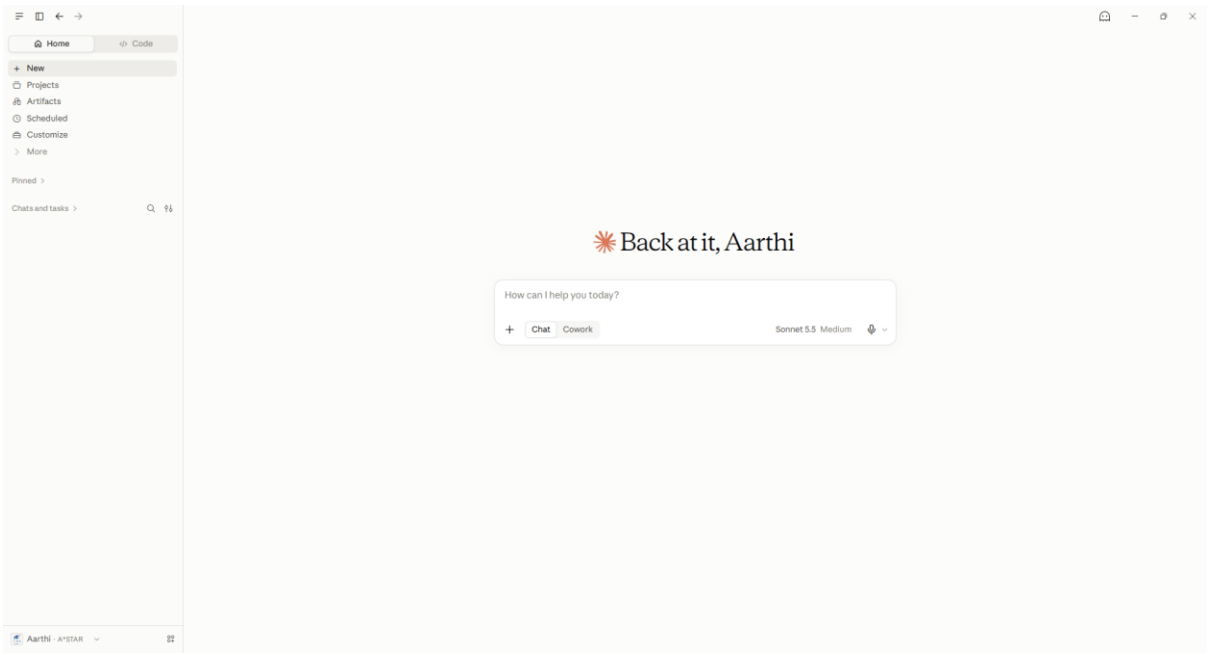
Co-AI BEGINNER WORKSHOP · AGENTIC AI FUNDAMENTALS
Aarthi Ravikrishnan & Deepa Ramasamy
14th October 2026

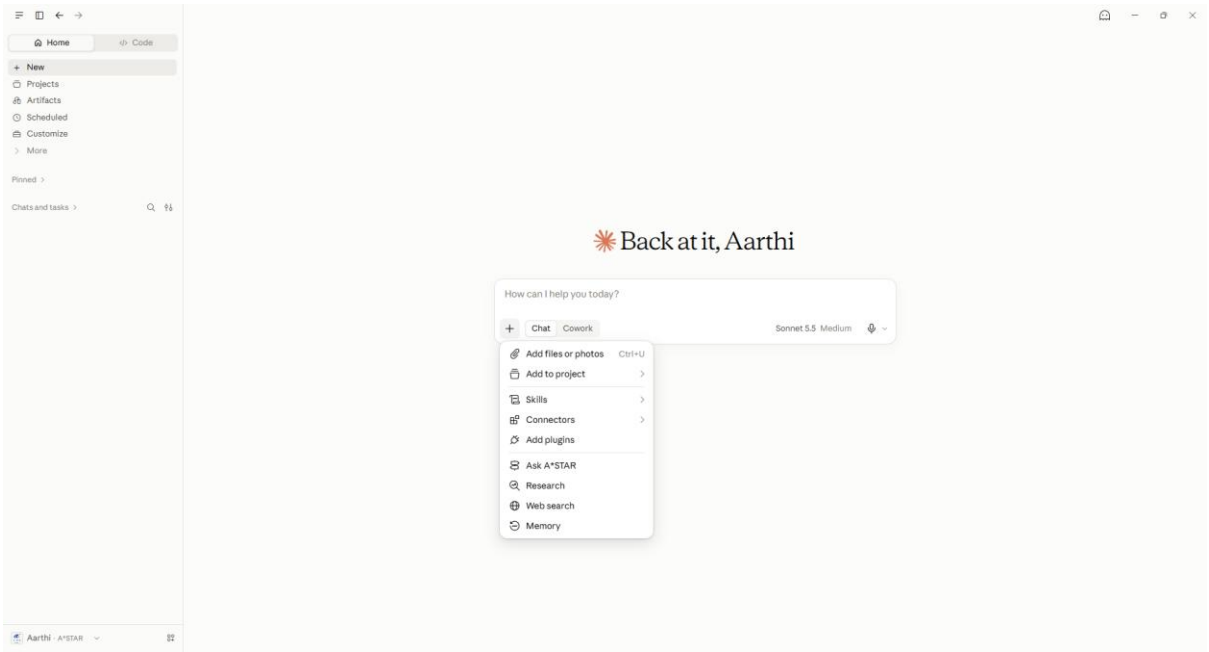
Example 1: Parsing messy clinical data into a structured CSV-

Tool: Claude (CHAT) · All data is synthetic

1. Navigate to `Example1_data_parsing/data` folder.
2. Inspect the files inside the folder by opening them separately.
    a. Are the entries structured?
3. Open Claude and you should see a window like this.



4. Upload the `files` in the data folder using the + option (`Add files or photos`) in the toolbox and upload the 3 files in `data` folder (opened in 1).



Naïve version

Example prompt

"Can you merge all these files and give a CSV file as output?"

Questions

1. What do you notice about the results?
2. Did it ask you for any inputs?

Beginner version

1. In the same chat, attach the file `skinad_instructions.md`, then send a short message such as

> Follow the attached SKIN-AD instructions and clean the three site files into clean_skinad.csv.

Questions

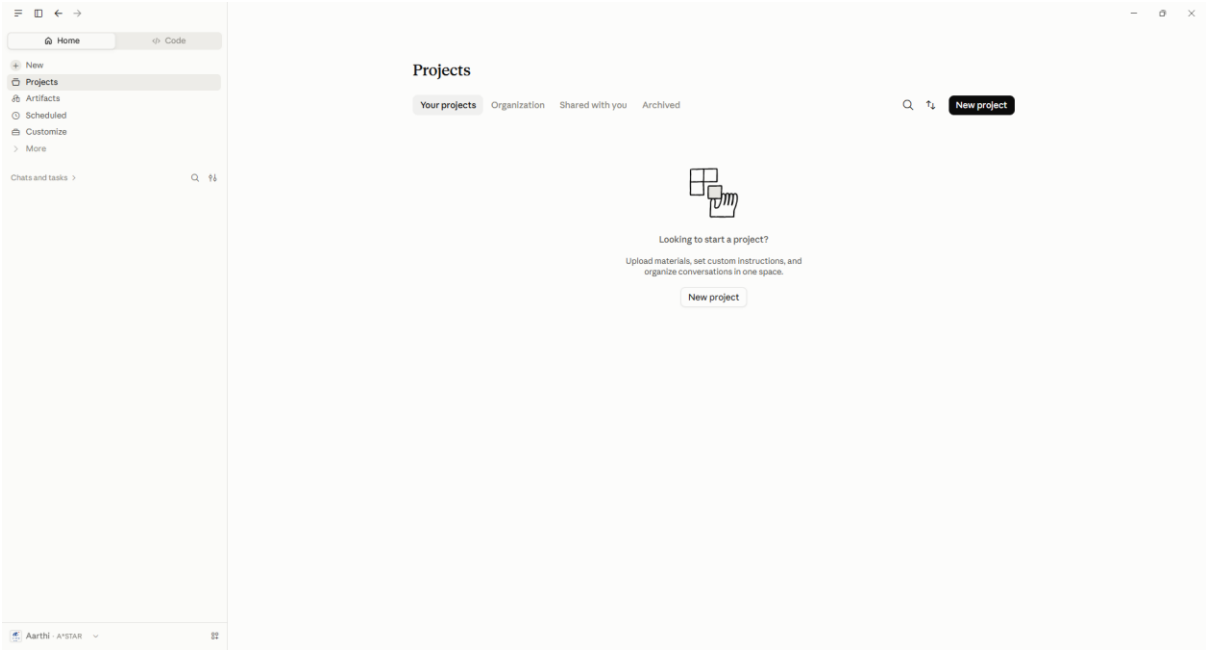
3. Do you notice any difference in way output is being produced?
4. Are there any differences in the output itself?

Intermediate version (1)

How to repeat this activity across multiple runs effectively?

Setup as a Claude Project

1. Open Claude and click Projects.

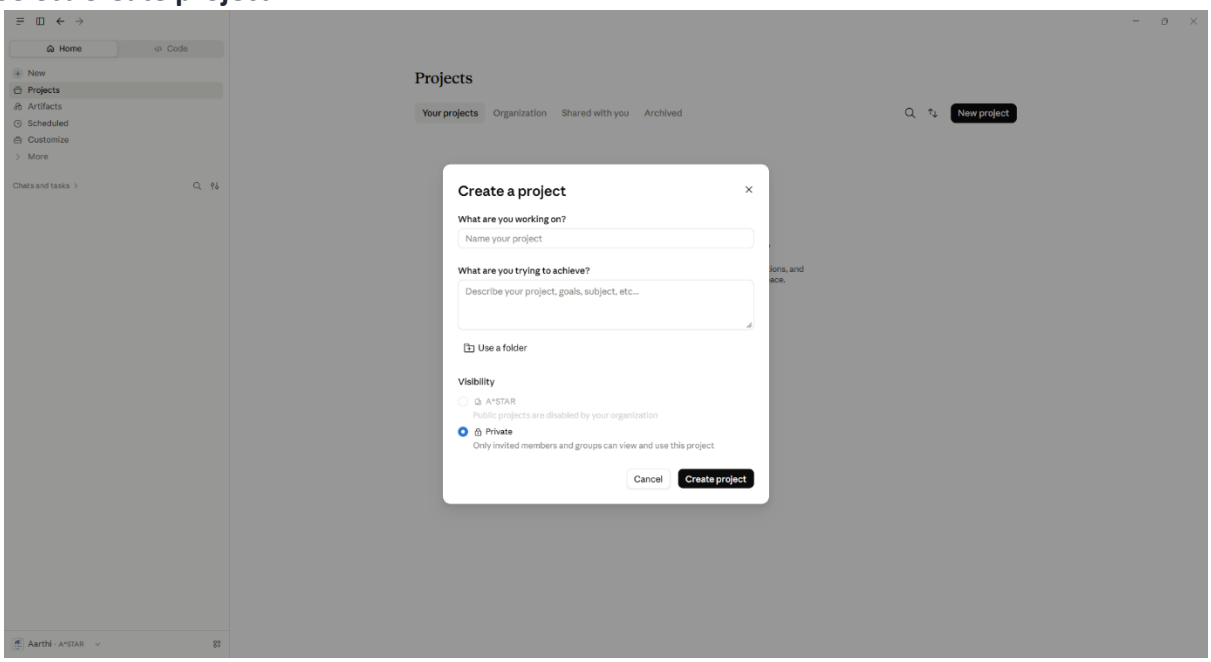


2. Enter the details:

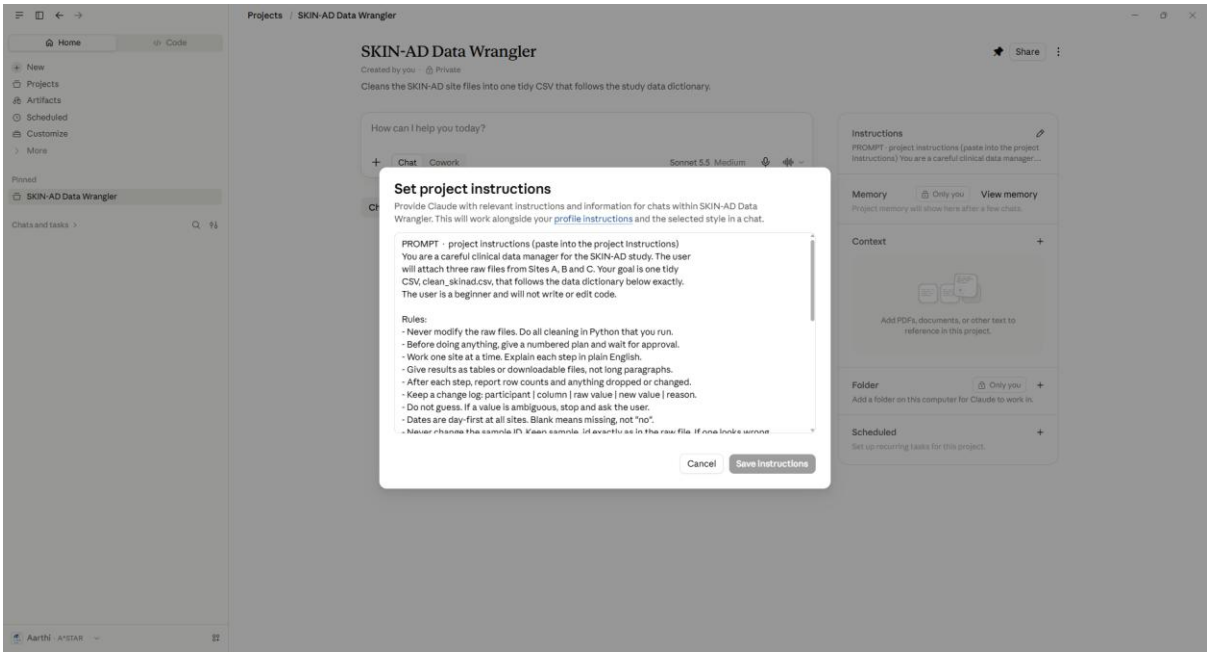
   Name: SKIN-AD Data Wrangler

   Description: Cleans the SKIN-AD site files into one tidy CSV that follows the study data dictionary.

   Select **Create project**.



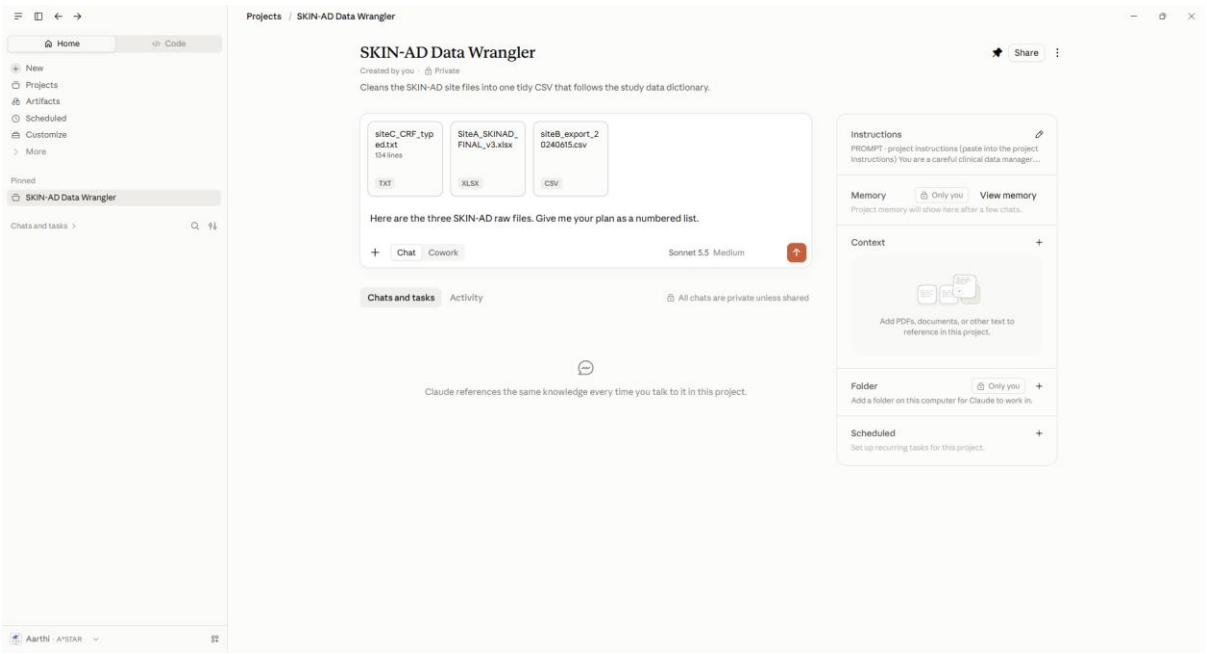
3. **Instructions:** on the project page, open **Instructions** (“Set project instructions”), paste the full text of the “project instructions” from the file “skinad_instructions.txt”.



4. Leave the other fields empty.

Start the work: open a new chat inside your project, attach the three files from data/ with the paperclip / + button, and send:

> **PROMPT · first message to Claude**
> `Here are the three SKIN-AD raw files. Give me your plan as a numbered list.`



How to carefully approach this problem of merging files?

Step 1 · Look before you code

> **PROMPT · profile the files**
> Before cleaning anything, profile each file. For each one, list every data-quality problem you can find: structure, headers, delimiters, encoding, codes, units, missing-value markers, duplicates, inconsistent spellings, and anything that is not stored as a plain cell value. Give me a table: file | problem | example | proposed fix.

Now do your own inspection in Excel and Notepad, and fill in the log below. Compare it with Claude's list. **Did you miss any?**

Step 2 · Site B: the "database" file

> **PROMPT · Site B**
> Start with siteB_export_20240615.csv. Load it and show me the first 10 rows as a table, plus a table of every column and its type (number, text or date). Then clean it into the target format. Read the lines at the top of the file before you start - they may tell you how values are coded.

Check: Are the numeric columns actually numbers, not text? Does every participant have the expected visits? Open the raw file in Notepad and compare two rows by eye.

Step 3 · Site A & C

Repeat the same with site A and C and check for how they compare with original files.

> **PROMPT · Site A**
> Now Site A (the .xlsx). Give me a table: sheet | what it contains. Tell me what information exists in this workbook that is NOT stored as a normal cell value, and how you will recover it. Then clean both visit sheets into the target format and combine them.

> **PROMPT · Site C**
> Site C is a text file of Key: value records, not a table. Split it into one record per participant and map each key to the target column, even when the same field is labelled differently. Show me the mapping as a table (raw label | target column), and list any records with a missing field.

Step 4: Combine all the three files

> **PROMPT · combine + validate**
>
> ```
> Combine the three sites into clean_skinad.csv, sorted by participant_id
> then visit. Then run these checks and show the results as a table
> (check | PASS/FAIL | details):
> - exactly the 14 columns, in the dictionary order
> - no duplicate participant_id + visit
> - every value inside its allowed set or range (incl. SCORAD 0-103)
> - plausible height, weight and BMI; visit 2 date after visit 1
> - withdrawn is identical on all of a participant's rows
> Give me clean_skinad.csv to download.
> ```

Now check it yourself, with no answer key. *At work nobody hands you the right answer*. These follow-up prompts work on any dataset:

> **PROMPT · check without an answer key**
>
> ```
> 1. Give me the change log as a CSV to download:
>    participant | column | raw value | new value | reason.
> 2. Make a table of how many rows and participants came from each file
>    and how many ended up in the output. Explain any difference.
> 3. Pick 3 random participants and show their raw values next to their
>    cleaned values, side by side.
> 4. Is there any information in these files you might have missed:
>    colours, comments, hidden columns or rows, extra sheets?
> ```

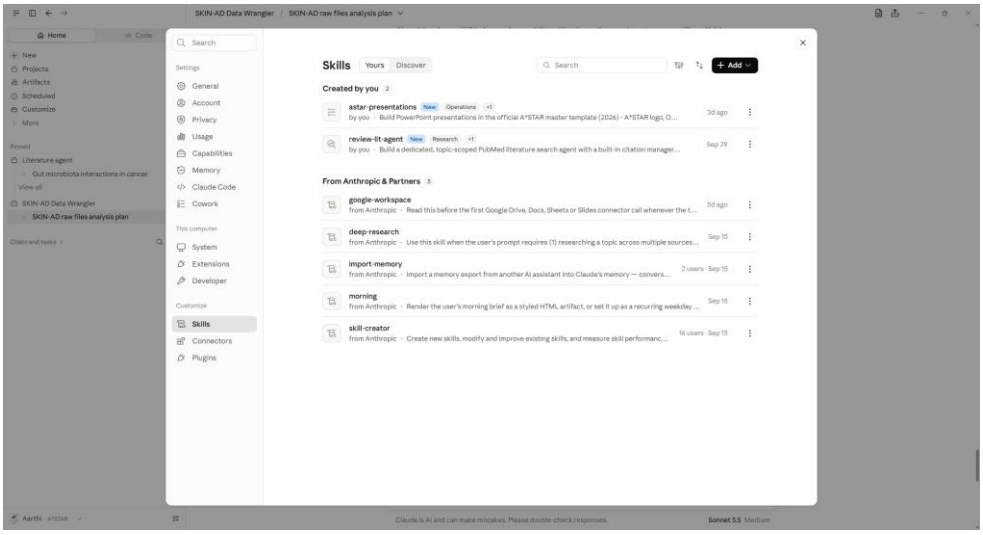
Check: Open the raw files and compare those 3 participants by eye. Read the change log: does every change make sense to you?

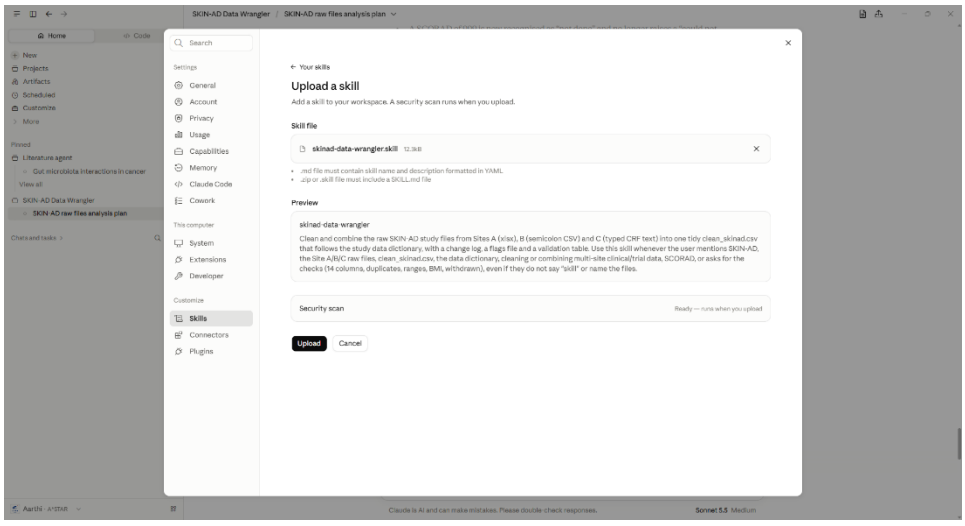
Making this reusable – SKILLS FILE

> **PROMPT ·**
> `Can you turn the instructions into a SKILLS file?`

Loading SKILLS into Claude on another machine

1. Navigate to Settings folder and add the skill files.





2. The Skills files should be accessible via the chat prompt. Now the exact set of instructions would be available even on the chat outside the project folder.


Home
Code
New
Projects
Artifacts
Scheduled
Customize
More
Pinned
Literature agent
Gut microbiota interactions in cancer
View all
SKIN-AD Data Wrangler
SKIN-AD raw files analysis plan
Chats and tasks
Aarthi · A*STAR
Add files
astar-presentations
deep-research
google-workspace
import-memory
morning
review-lit-agent
skill-creator
skinad-data-wrangler
/Type to filter
Chat
Cowork
ernoon, Aarthi
Clean and combine the raw SKIN-AD study files from Sites A (xlsx), B (semicolon CSV) and C (typed CRF text) into one tidy clean_skinad.csv that follows the study data dictionary, with a change log, a flags file and a validation table. Use this skill whenever the user mentions SKIN-AD, the Site A/B/C raw files, clean_skinad.csv, the data dictionary, cleaning or combining multi-site clinical/trial data, SCORAD, or asks for the checks (14 columns, duplicates, ranges, BMI, withdrawn), even if they do not say "skill" or name the files.
skinad-data-wrangler is ready to use.

How to make a SKILL file on Claude?

